FROM GOOGLE AI STUDIO POC TO ANTIGRAVITY ENTERPRISE RC

Assurance, Self-Auditing & GenAI FinOps Economics

How **AegisAI Enterprise** was prototyped in **Google AI Studio (Gemini 3.6 Flash)**, architected with **Gemini 3.1 Pro (5%)**, engineered at scale in **Google Antigravity (Gemini 3.8 Flash — 80% combined Flash)**, self-audited (68.9% → 97.8%), hardened & QA-verified with **Gemini 4.0 Argon (10%)** and **Claude Opus 5.5 Max (5%)**, and presented as a Release Candidate to **Farmacias del Ahorro**, **Asociación Mexicana de Franquicias**, and **Grupo Coppel**—for a total build cost of **$215.24 USD** ($319.80 w/ full agentic loops).

1. AI STUDIO POC (GEMINI 3.6 FLASH)
2. ARCH (3.1 PRO 5%) + BULK BUILD (3.8 FLASH 74%)
3. HARDENING & QA (4.0 ARGON 10% + OPUS 5.5 5%)
4. RC: FARMACIAS DEL AHORRO • AMF • COPPEL
5. AUDIT COST: $0.43–$1.55 (BATCH: $0.21) VS $8.55–$31.09
6. TOTAL APP BUILD COST: $215.24 USD

END-TO-END MULTI-MODEL ENGINEERING LIFECYCLE

How We Built, Self-Audited, and Validated AegisAI

A four-stage right-sized multi-model engineering journey (80% Flash • 10% Argon • 5% 3.1 Pro • 5% Opus 5.5 Max).

PHASE 1 • PROOF OF CONCEPT (~6% TOKENS)

1. Prototyped in Google AI Studio (Gemini 3.6 Flash — $6.53)

• Built the initial Proof of Concept in **Google AI Studio** using **Gemini 3.6 Flash** (~45 iterations, 4.20M In / 0.90M Out tokens @ official $0.75/$3.75 rate).
• Validated structured JSON Schema output generation, multimodal PDF/diagram ingestion, and initial React UI scaffolding.
• Proved that Flash-class models with strict system instructions and temperature `0.0` evaluate complex security controls rapidly.

PHASE 2 • ARCHITECTURE (5%) & BULK BUILD (74%)

2. Architected in 3.1 Pro ($14.06) & Built in 3.8 Flash ($70.65)

• **Architecture (Gemini 3.1 Pro, 5%):** Designed the 4-layer Zero-Trust topology, 12-batch Map-Reduce engine, 265-control taxonomy & Firestore security model (3.55M In / 0.58M Out @ $2.00/$12.00).
• **Bulk Construction (Gemini 3.8 Flash, 74%):** Executed **171+ engineering prompts** in **Google Antigravity** to build the Express/React 19 stack, GCP Live Scanner, 17 bilingual PDF manuals ($65.85), and live Vertex AI self-audits ($4.80). Combined Flash = **80% of tokens for $77.18** ($38.59 Batch).

PHASE 3 • SELF-AUDIT, HARDENING (10%) & QA (5%)

3. Self-Audited (68.9%➔97.8%) + Hardened in Argon & Opus 5.5

• Baseline self-audit (`MULTI-2026-0437`) on `Originaldocs.zip` scored **68.9%** (35 gaps).
• **Gemini 4.0 Argon (10%, $25.70):** Keyless Vertex AI ADC, Cloud IAP perimeter, 308 Firestore rules unit tests & 4K/8K architecture diagrams (7.10M In / 1.15M Out @ $2.00/$10.00).
• **Claude Opus 5.5 Max (5%, $96.45):** Adversarial red-team review, constant-time auth, SSRF/DNS rebinding fixes & QA ➔ **97.8%** (`MULTI-2026-8983` , 0 failures).

PHASE 4 • ENTERPRISE FIELD VALIDATION

4. Release Candidate (RC) Presented to Industry Leaders

• Presented the hardened Release Candidate (RC) to three major Mexican organizations:
1. **Farmacias del Ahorro** (Healthcare & Retail Pharmacy)
2. **Asociación Mexicana de Franquicias (AMF)** (Franchise Ecosystem)
3. **Grupo Coppel** (Financial Services & Omnichannel Retail)

• Validated multi-standard compliance (OWASP AI-SVS, ISO 42001, LFPC & LFPDPPP).

EATING OUR OWN DOGFOOD

Running AegisAI's Codebase & Documentation Against Itself

How AegisAI diagnosed 35 OWASP AI-SVS & ISO 42001 gaps in its own prototype and guided its own remediation to 97.8%.

1. Baseline Self-Audit (`MULTI-2026-0437`)

- **Overall Score: 68.9% (Level L2)**
- **OWASP AI-SVS v1.0:** 71.8% (17 gaps)
- **ISO/IEC 42001:2023:** 66.7% (18 gaps)
- **Key Findings Detected by AegisAI:**
  - Writable `/app/config` in container (`C04.1`)
  - Timing-vulnerable token check `===` (`C05.3`)
  - Legacy `roles/viewer` on runtime SA (`C05.2`)
  - Missing SHA-256 normative manifest (`C01.4`)
  - Missing ISO 42001 AIMS Policy & Appeal docs

2. Full-Stack Remediation in Antigravity

- **Generator Fixes:** Untruncated Section 4 Cookbook (all 35 patches), smart sentence-boundary splitter, accurate syntax tags.
- **Container & Cloud:** `chmod -R a-w /app` (`C04.1`), Cloud IAP + SSRF/IMDS blocker (`C04.3`), pruned `roles/viewer` (`C05.2`).
- **Runtime Guardrails:** SHA-256 manifest (`C01.4`), crescendo detector (`C02.10`), `crypto.timingSafeEqual` (`C05.3`), Anti-DoW quotas (`C11.1`), sagas (`C09.4`).
- **Docs & Rules:** Authored Manual 8 (ISO 42001 AIMS) & passed 308/308 Firestore rules unit tests.

3. Post-Fix Re-Audit (`MULTI-2026-8983`)

- **Overall Score: 97.8% (+28.9 pts)**
- **ISO/IEC 42001:2023: 100.0%** (54/54 mandatory passed, **0 Failed**)
- **OWASP AI-SVS v1.0: 94.7%** (45/49 passed, **0 Failed**)
- **Telemetry Recorded in Firestore:**
  - Model: `gemini-3.8-flash` on Vertex AI
  - Input Tokens: `1,758,724`
  - Output Tokens: `100,715`
  - Zero critical or high-risk failures remaining

Why Self-Auditing Matters: Running AegisAI against its own architecture and manuals proved that the evaluator does not hallucinate approvals—it caught subtle issues (such as `providedToken === configuredToken` and missing ISO 42001 whistleblowing/appeal channels) and verified their closure once remediated.

MARKET VALIDATION & EXECUTIVE ROADSHOW

Release Candidate (RC) Presented to Strategic Enterprises

Following our self-audit hardening loop, the AegisAI Release Candidate was presented to three flagship Mexican organizations.

HEALTHCARE & RETAIL PHARMACY

Farmacias del Ahorro

- **Use Case Focus:** Auditing clinical/pharmacy digital assistants, prescription triage bots, and omnichannel customer service AI.
- **Key AegisAI Capabilities Demonstrated:**
  - Strict **LFPDPPP** sensitive health/PII data protection & pre-inference DLP masking.
  - Hallucination prevention (`temperature: 0.0`) and human escalation workflows (`ISO 42001 A.5.2`).
  - Continuous RAG grounding telemetry (`C08.4`) on Google Cloud Vertex AI.

NATIONAL FRANCHISE ECOSYSTEM

Asociación Mexicana de Franquicias

- **Use Case Focus:** Standardizing AI adoption, customer chatbots, and automated marketing compliance across hundreds of member franchises.
- **Key AegisAI Capabilities Demonstrated:**
  - **LFPC (Profeco 2026)** compliance: preventing deceptive AI commercial patterns and enforcing Art. 56 5-day cooling-off retraction.
  - **BYOF Engine:** Custom franchise governance templates with low-cost Flash evaluations (`$0.43/audit` standard or `$0.21/audit` Batch).

FINANCIAL SERVICES & RETAIL

Grupo Coppel

- **Use Case Focus:** High-scale credit scoring ML, fraud detection engines, and enterprise GenAI engineering governance.
- **Key AegisAI Capabilities Demonstrated:**
  - Simultaneous **OWASP AI-SVS v1.0 (Level L2/L3) + ISO/IEC 42001:2023** certification readiness.
  - Multi-tenant domain isolation (`C05.4` , `C08.1`), Cloud IAP Zero-Trust perimeter, and keyless **GCP Live Scanner**.
  - Drop-in remediated architecture & Terraform/TypeScript cookbook generation.

Executive Feedback Highlights: Enterprise stakeholders highlighted AegisAI's ability to combine **global technical standards (OWASP AI-SVS + ISO 42001)** with **Mexican statutory law (LFPDPPP + LFPC)** while delivering ready-to-deploy remediation code at a fraction of traditional consulting costs.

FINOPS ANALYSIS: COST PER COMPLETE OWASP AI-SVS + ISO/IEC 42001 AUDIT

Gemini (3.8 Flash, 4.0 Argon, 3.1 Pro) vs. Claude (Fable 5.1, Opus 5.5)

Calculated from real production Firestore token telemetry using Official 2026 Rate Cards per full dual-standard audit.

STANDARD AUDIT (3.8 FLASH)
$0.43 USD — 1.0X BASE
Batch/Flex Tier: $0.21 USD

10-PDF SUITE AUDIT (3.8 FLASH)
$1.55 USD — 12 BATCHES
1.64M In + 86.5K Out ($0.78 Batch)

COST ADVANTAGE VS. OPUS 5.5
20.0x Cheaper — VERIFIED
40.0x vs Batch • 6.7x vs Std Opus

MONTHLY OPEX SAVINGS (1K AUDITS)
95.0% Saved — -$8,122/MO
$428/mo vs. $8,550/mo (Opus Max)

Per-Audit Cost & Multiplier Benchmark (USD)

Dual-Bar Comparison: Standard Audit (370K In / 40K Out) vs. Full 10-PDF Suite (1.64M In / 86.5K Out)

Standard Audit · 10-PDF Bundle

Gemini 3.8 Flash DEFAULT ENGINE $0.75 In / $3.75 Out — 1.0X BASELINE
Std: $0.43 (Batch $0.21)
10-PDF: $1.55 ($0.78)

Gemini 4.0 Argon FRONTIER $2.00 In / $10.00 Out — 2.67X VS FLASH
Std: $1.14 USD
10-PDF: $4.15 USD

Gemini 3.1 Pro DEEP REASONING $2.00 In / $12.00 Out — 2.78X – 2.85X
Std: $1.22 USD
10-PDF: $4.32 USD

Claude Fable 5.1 Medium EXTERNAL $3.00 In / $15.00 Out — 4.0X (8.0X VS BATCH)
Std: $1.71 USD
10-PDF: $6.22 USD

Claude Opus 5.5 Max MAX TIER $15.00 / $75.00 (Std: $5/$25) — 20.0X (40X BATCH / 6.7X STD)
Std: $8.55 (Std $2.85)
10-PDF: $31.09 ($10.36)

Monthly Run-Rate @ 1,000 Audits / Mo

Total monthly cloud inference bill for 1,000 standard dual-standard audits

SAVE $8,122 / MO



Production Token Telemetry (Cloud Firestore)

15 Map-Reduce Calls / Audit

Profile A: Standard Architecture Audit (408.6K tok) — 90.3% In / 9.7% Out

Audits MULTI-2026-0507 & 1250 : Avg 368,827 Input + 39,812 Output tokens

Profile B: Full 10-PDF Suite Audit (1.725M tok) — 95.0% In / 5.0% Out

Audits MULTI-2026-0437 & 8983 : Avg 1,638,410 Input + 86,495 Output (~136K ctx/batch)

Telemetry Source: Cloud Firestore (aegis-ai-db) Audits MULTI-2026-0437, 8983, 0507 & 1250

ENGINEERING ROI & MULTI-MODEL LIFECYCLE BREAKDOWN (80% FLASH • 10% ARGON • 5% PRO • 5% OPUS)

How Much Money We Spent Creating AegisAI: $215.24 USD Total

Right-sized multi-model engineering across AI Studio PoC, System Architecture, Bulk Full-Stack Construction, Frontier Hardening & Adversarial QA.

$215.24
Total Money Spent creating AegisAI ($319.80 w/ full agentic loops)

~71.0M
Total Input Tokens (80% Flash for $77.18 • 10% Argon • 5% Pro • 5% Opus)

$1,931
What the exact same token volume costs if run 100% on Claude Opus 5.5 Max

~$95,000
Equivalent traditional engineering cost (4 DevSecOps engineers × 3 months)

Development Phase & Role	Model & Token Share	Engineering Scope & Token Volume	Actual Spend (2026)	If 100% Opus 5.5
1. Rapid PoC & Schema Validation	AI Studio • `Gemini 3.6 Flash` (~6%)	~45 PoC iterations, JSON schema & PDF tests • 4.20M In / 0.90M Out	**$6.53 USD**	$130.50 USD
2. System Architecture & Blueprint	Antigravity • `Gemini 3.1 Pro` (5%)	Zero-Trust topology, 12-batch Map-Reduce, 265 controls • 3.55M In / 0.58M Out	**$14.06 USD**	$96.45 USD
3. Bulk Build, Manuals & Audits	Antigravity + Vertex • `Gemini 3.8 Flash` (74%)	171 prompts, 17 PDFs ($65.85) + Live Self-Audits ($4.80) • 52.60M In / 8.32M Out	**$70.65 USD**	$1,412.70 USD
4. Frontier Hardening & Visuals	Antigravity • `Gemini 4.0 Argon` (10%)	Keyless Vertex ADC, Cloud IAP, 308 rules tests, 4K diagrams • 7.10M In / 1.15M Out	**$25.70 USD**	$192.90 USD
5. Adversarial Bugfixing & QA	Antigravity • `Claude Opus 5.5 Max` (5%)	Red-team code review, timing-safe auth, SSRF/DNS fixes • 3.55M In / 0.58M Out	**$96.45 USD**	$96.45 USD
6. Google Cloud Serverless Infra	Cloud Build (9x), Cloud Run, Firestore, IAP	9 container builds, scale-to-zero runtime, `aegis-ai-db` • **Total: $215.24 USD**	**$1.85 USD**	$1.85 USD

THE PROBLEM TO SOLVE

The Critical Security Blindspot in Enterprise GenAI Adoption

Enterprises are shipping generative AI models, agentic workflows, MCP tools, and RAG pipelines without architectural verification or ISO 42001 governance.

⚠ Ungoverned AI Access & Breaches

Per **IBM Cost of a Data Breach 2025**, **13% of organizations** reported breaches of AI models or applications, and **97% of those** lacked proper AI access controls or formal impact assessments (`ISO 42001 A.5.1`).

⚠ Multi-Turn & Agentic Attack Surface

RAG pipelines, MCP servers, and autonomous agents introduce critical vectors: **Multi-Turn Crescendo Prompt Injection** (`C02.10`), **Normative/Vector Poisoning** (`C01.4`), and SSRF/IMDS exfiltration (`C04.3`).

⚠ CISO vs Data Science Silo

Security teams lack tools to audit agent graphs, MCP manifests (`C10.1`), and pre-retrieval vector ACLs (`C08.1`); AI developers lack automated mappings to formal compliance clauses.

Lack of Operationalized Governance (ISO/IEC 42001)

Organizations struggle to operationalize **ISO/IEC 42001:2023** requirements—such as Corporate AI Policies (`A.2.1`), Confidential Whistleblowing (`A.3.3`), AI Asset Inventories (`A.4.1`), and Human Verdict Appeals (`A.5.2`)—into software engineering workflows.

Proliferation of "Shadow AI" & Denial of Wallet

Unregulated external API calls, timing-vulnerable token checks (`C05.3`), over-privileged service accounts (`C05.2`), missing Anti-DoW token budgets (`C11.1`), and exposed logit probabilities (`C11.4`).

Source: OWASP GenAI Security Project (AI-SVS v1.0) & ISO/IEC 42001:2023

COMPLIANCE MATRIX

Multi-Standard Framework Support & ISO/IEC 42001 AIMS

Simultaneous evaluation across global technical security standards, management systems, and regional privacy statutes.

Standard / Framework	Technical & Governance Scope	Key Controls Verified & Enforced by AegisAI	Assurance Level
OWASP AI-SVS v1.0	AI Security Verification Standard (12 Chapters)	`C01.4` SHA-256 Manifest, `C02.10` Crescendo Guard, `C04.1` Read-Only Container, `C04.3` Cloud IAP & SSRF Block, `C05.2–C05.4` Constant-Time Auth & IAM, `C08.1/C08.4` RAG ACLs, `C09.4` Sagas, `C10.1–C10.5` MCP Security, `C11.1/C11.4` Anti-DoW & Logprobs Off, `C12.2` SIEM Alerts.	L1 / L2 / L3 READY
ISO/IEC 42001:2023	AI Management System (AIMS Clauses 4–10 & Annex A)	`Clause 4.1/4.2` Context & Stakeholders, `Clause 5/A.2.1/A.2.3` Corporate AI Policy, `Clause 7/A.3.3` Competency & Whistleblowing, `A.4.1` AI Asset Inventory, `A.5.1–A.5.3` Impact Assessment & Human Appeal, `Clause 9` KPIs & Internal Audit, `A.9.1/A.9.2` Acceptable Use.	CERTIFICATION READY
Mexican Regulatory (2026)	Consumer Protection (LFPC) & Personal Data (LFPDPPP)	Algorithmic transparency, deceptive commercial pattern prevention, Art. 56 5-day cooling-off retraction, ARCO cascading deletion, and DLP PII masking.	REGIONAL STATUTORY
BYOF Custom Framework	*Bring Your Own Framework* (Multi-Tenant)	Admin-authenticated ingestion (`C05.3`) with SHA-256 integrity checks (`C01.4`) and pre-retrieval tenant domain ACL filtering (`allowedDomains` , `C08.1`).	TENANT ISOLATED

Unified Cross-Framework Map-Reduce: Evaluates 103+ combined OWASP AI-SVS & ISO/IEC 42001 controls in parallel batches using constrained chapter ID enums so ISO Clauses 9/10 and Annex A.9/A.10 never collide with OWASP C09/C10.

SOFTWARE ARCHITECTURE

4-Layer Zero-Trust & Defense-in-Depth Architecture

Hardened serverless architecture deployed on Google Cloud Run behind Identity-Aware Proxy (IAP) with keyless Vertex AI inference.

1. Perimeter & SPA

- **Cloud IAP** (`--no-allow-unauthenticated`)
- **React 19 + TS + Vite**
- Firebase Auth OAuth 2.0 domain allowlist (`google.com` , `altostrat.com`)
- Embedded ISO 42001 Policy, Appeal (`A.5.2`) & Whistleblowing (`A.3.3`) UI

2. Hardened API Layer

- **Read-Only Container** (`chmod -R a-w /app` , `C04.1`)
- Constant-time token check (`crypto.timingSafeEqual` , `C05.3`)
- Firebase Admin `verifyIdToken()`
- DNS Rebinding Guard (`C10.5`) & MCP Signature Verifier (`C10.1`)
- Anti-DoW Quotas (`C11.1`)

3. AI Engine & Guardrails

- **Vertex AI** (`gemini-3.8-flash`) via keyless ADC
- Least-privilege SA (`aegis-runner-sa` : zero `roles/viewer` , `C05.2`)
- Multi-Turn Crescendo Detector (`C02.10`)
- `responseLogprobs: false` (`C11.4`)
- Outbound SSRF/IMDS Block (`C04.3`)

4. Data, Integrity & Sagas

- **Cloud Firestore** (`aegis-ai-db`) via Admin SDK
- 308/308 tested `firestore.rules`
- Immutable audits (`allow delete: if false`) & `ownerUid` ACLs (`C05.4`)
- SHA-256 Manifest (`C01.4`)
- Compensating Sagas (`C09.4`)

Zero-Trust Isolation & Structured Security Alerting (C12.2):

Runtime state writes exclusively to Firestore (`system_config/admin_settings`) and ephemeral `/tmp/aegis-runtime-cache` . Every security event (prompt injection escalation, SSRF block, token mismatch, DoW quota breach, saga rollback) emits a structured JSON alert (`aegis-security-alerting-c12-2`) to Google Cloud Logging.

OPERATIONAL EXECUTION

The Auditor's Workflow: From Ingestion to 100% Hardening

Six-stage deterministic assurance lifecycle with built-in human oversight and automated code remediation.

1 Authenticated Ingestion & Pre-Flight Guardrails
Corporate user (verified via Cloud IAP + Firebase ID Token) uploads PDFs/ZIPs/specs; `inspectPromptAndTrackSession` (`C02.10`) and DLP masking sanitize input.

2 SHA-256 Verified & Tenant-Filtered Context
Normative snapshots verified against `integrity-manifest.json` (`C01.4`) and custom frameworks filtered by `tenantDomain` (`C08.1`) are loaded.

3 Deterministic Vertex AI Map-Reduce Evaluation
`gemini-3.8-flash` (`temperature: 0.0` , `responseLogprobs: false` , 180s timeout) evaluates each batch under strict chapter enum schemas.

4 Executive Dashboard & Human Appeal (ISO A.5.2)
Interactive radar charts, L1/L2/L3 maturity breakdown, and direct Verdict Appeal / Human Review workflow (`ai-appeals@aegisai.enterprise`).

5 100% Remediated Architecture & Full Cookbook
Synthesizes Section 3 drop-in architecture + Section 4 Cookbook for all `1..N` findings with accurate syntax tags (`typescript` , `hcl` , `yaml` , `python` , `bash`).

6 Immutable Persistence & Compensating Saga
Saved to Firestore with `ownerUid` / `tenantDomain` attribution (`C05.4`), `runCompensatingSaga` rollback safety (`C09.4`), and AIMS KPI recording (`C08.4`).

END-TO-END REMEDIATION TRACEABILITY

OWASP AI-SVS v1.0 & ISO/IEC 42001:2023 Controls Enforced

Comprehensive architectural and governance controls implemented across code, container, cloud perimeter, and AIMS policies.

OWASP AI-SVS v1.0 Technical Controls (17 Hardened Domains)

- **C01.4 Normative Integrity:** `verifyNormativeSnapshotsIntegrity()` SHA-256 manifest check at startup.
- **C02.10 Crescendo Defense:** `inspectPromptAndTrackSession()` multi-turn escalation blocker (HTTP 422).
- **C04.1 & C04.3 Infrastructure:** Read-only `/app` (`chmod -R a-w`), Cloud IAP, VPC egress & `validateOutboundEndpoint()` SSRF/IMDS block.
- **C05.2, C05.3 & C05.4 Auth/IAM:** Pruned `roles/viewer` on `aegis-runner-sa` , `crypto.timingSafeEqual` , Firebase ID token verification & Firestore `ownerUid` ACLs.
- **C08.1 & C08.4 RAG/Grounding:** Tenant domain ACL pre-filtering & continuous grounding telemetry.
- **C09.4 & C10.1–C10.5 Agents/MCP:** `runCompensatingSaga()` rollback, MCP Ed25519/HMAC manifest verification, OAuth 2.1 PKCE & `dnsRebindingGuard` .
- **C11.1, C11.4 & C12.2 FinOps/SIEM:** `acquireAntiDoWQuota()` , `responseLogprobs: false` & `emitSecurityAlert()` .

ISO/IEC 42001:2023 AIMS Governance Controls (18 Clauses & Annex A)

- **Clause 4.1 & 4.2 Context & Stakeholders:** Formal internal/external issue matrix, societal role as AI assurance provider & stakeholder mapping.
- **Clause 5.1, 5.2, A.2.1 & A.2.3 Leadership & AI Policy:** Executive-approved Corporate AI Policy with annual revision cycle.
- **Clause 7.2, 7.3 & A.3.3 Competency & Whistleblowing:** Mandatory role training & confidential ethics reporting (`ai-ethics-whistleblower@aegisai.enterprise`).
- **A.4.1 Resource & Tooling Inventory:** Cataloged models, normative corpora, runtime SAs & KMS secrets.
- **A.5.1, A.5.2 & A.5.3 Impact Assessment & Appeals:** Formal AIIA methodology, quantitative risk thresholds & human appeal channel (`ai-appeals@aegisai.enterprise`).
- **Clause 9.1–9.3 & A.9.1/A.9.2 Assurance & Use:** Continuous KPIs, semi-annual internal audits, management review & acceptable use notices.

EVOLUTION & ROADMAP

Next Horizons in AegisAI Platform Engineering

Our forward-looking roadmap following our Release Candidate presentations to Farmacias del Ahorro, AMF, and Grupo Coppel.

Q4 2026 • NEAR TERM

Native CI/CD Shift-Left Gates

Official **GitHub Actions** and **GitLab CI Runners** to automatically block pull requests introducing prompt injection vulnerabilities, unsigned MCP manifests, or unencrypted vector storage before merging.

Q1 2027 • MEDIUM TERM

Multi-Agent Graph Auditing

Specialized auditing module for complex autonomous agent workflows (LangGraph, CrewAI, AutoGen): tool authorization boundary checks, compensating saga verification, and inter-agent memory sandboxing.

Q2 2027 • LONG TERM

Autonomous IaC Remediation

One-click automated synthesis of **Terraform Modules** and **Helm Charts** that directly apply Cloud IAP, VPC Service Controls, and Cloud Armor remediation recipes to cloud clusters.

🎯 Conclusion: Enterprise AI Assurance Built for $215.24 Multi-Model, Auditing for $0.43–$1.55 (Batch: $0.21)

By combining **Gemini 3.6 & 3.8 Flash (80%)** for rapid PoC and bulk construction with **Gemini 3.1 Pro (5%)** for architecture and **Gemini 4.0 Argon (10%) + Claude Opus 5.5 Max (5%)** for deep hardening & QA, AegisAI was built at **88.9% lower cost than all-Opus ($1,931)** and **99.77% lower cost than traditional staffing (~$95k)**.

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